UNVEILING ANTIMICROBIAL PEPTIDE INTERACTIONS: FROM VESICLES TO LIVING CELLS VIA QUANTITATIVE SPECTROSCOPIC MEASUREMENTS

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ABSTRACT

Antimicrobial peptides (AMPs) are promising candidates to combat the global rise of antibiotic-resistant bacteria. The effectiveness of AMPs depends on their affinity for lipid bilayers and their ability to perturb membrane integrity upon binding. Traditionally, their activity and selectivity toward pathogens are assessed using microbiological assays or model vesicles. However, these parameters are typically evaluated in isolated systems containing only a single cell or vesicle type at a time, and peptide membrane affinity, a key determinant of selectivity, is often neglected. Consequently, these models fail to accurately reflect *in vivo* conditions. Here, we investigate AMP selectivity in mixed environments using model membrane systems designed to better mimic the complex conditions occurring during bacterial infections. In addition, we extend our previously developed method to the quantitative evaluation of AMP–cell interactions in living eukaryotic cells. This approach relies on spectroscopic techniques under experimental conditions closely resembling those of microbiological assays, thereby bridging the gap between biophysical and microbiological studies. We examined peptides binding to a murine macrophage cancer cell line and healthy Chinese hamster ovary (CHO) cells, also employing confocal microscopy to gain deeper insight into the peptide–cell interaction.

We demonstrate that AMP interactions with living eukaryotic cells can be quantitatively assessed, thereby elucidating the mechanisms of their selectivity.